

1.System Overview

- 1.1 Architecture
- 1.2 Execution Workflow
- 1.3 Design Principles

2. Execution Stages

- 2.1 Initialization
- 2.2 Planning
- 2.3 Tool Execution

2.4 Main Loop (Repeat)

- 2.5 Observation
- 2.6 Termination (Exit)

3. State and Data

- 3.1 Key States
- 3.2 Data Flow

4. Components

- 4.1 Prompting
- 4.2 Memory
- 4.3 Tools
- 4.4 Others
- ...

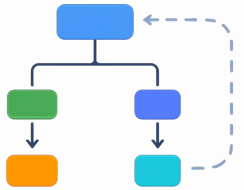
2.4 Main Loop

The core orchestrator that iteratively performs reasoning, tool use, and observation until the exit conditions are met

- Purpose**
Drive the agent to iteratively complete the task >
- Triggering Conditions**
Enter when loop starts or after each observation >
- Input**
Current state, observation, memory, tool results >
- Processing Steps**
Reasoning, tool selection, action execution, observation >
- Output**
Updated state and observation for next iteration >
- Related States**
Messages, scratchpad, tool results, loop counter >
- Exceptional Cases**
Tool failure, parse error, max steps reached, timeout >
- Key Functions**
_run_agent_loop(), _step(), _should_exit() >
- Configurations**
Max steps, retry policy, exit criteria >
- Examples**
Typical execution trace and logs >
- Notes**
Loop termination depends on multiple exit conditions >

L1

Level 1:
System Overview



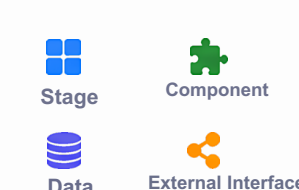
Overall description of the entire harness system, including architecture, execution model, major stages, and system goals.

Information Included

- ✓ System architecture
- ✓ Execution model
- ✓ Stage overview
- ✓ Data & state flow
- ✓ Design principles
- ✓ Entry and exit conditions

L2

Level 2:
Component Overview



Overview of all working units that constitute the harness, including their roles, inputs/outputs, and interactions.

Information Included

- ✓ Component list & taxonomy
- ✓ Responsibilities & goals
- ✓ Inputs / outputs
- ✓ Dependencies & interactions
- ✓ Key states involved

L3

Level 3:
Unit Deep Dive



Detailed analysis of each unit's internal logic, state transitions, edge cases, and implementation.

Information Included

- ✓ Detailed execution logic
- ✓ State transitions
- ✓ Exception handling
- ✓ Implementation details
- ✓ Source code & snippets
- ✓ Logs & examples

Shallow(High-level)

Deep (Implementation)